

ITELEC 4100
Lab # 2

INTRO TO AJAX

1

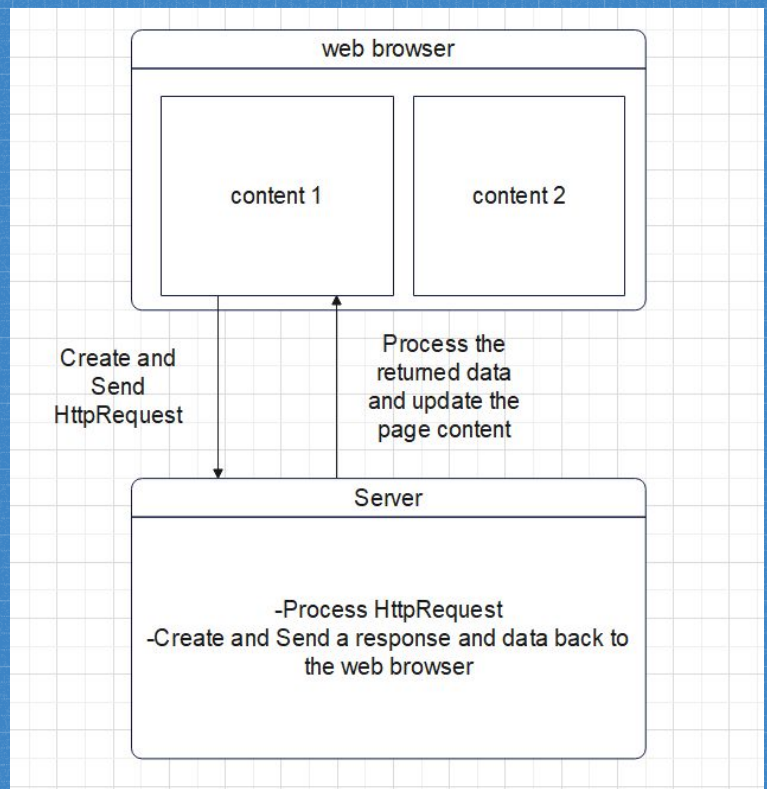
INTRODUCTION TO AJAX

What is AJAX?

- AJAX or Asynchronous JavaScript and XML enable us to load data from the server without refreshing the whole web page

3

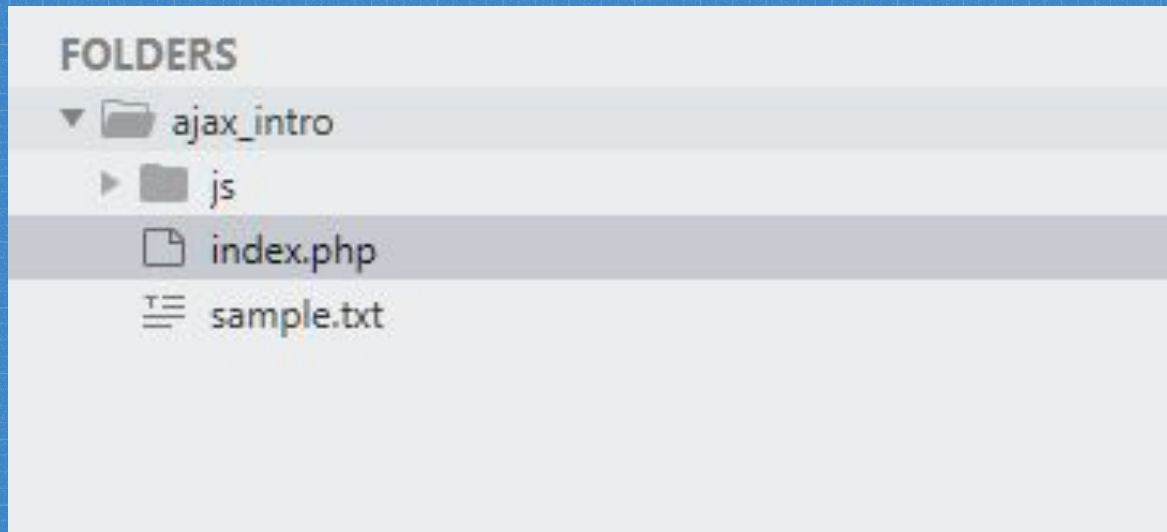
AJAX Concept



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AJAX using Vanilla JavaScript (exercise)

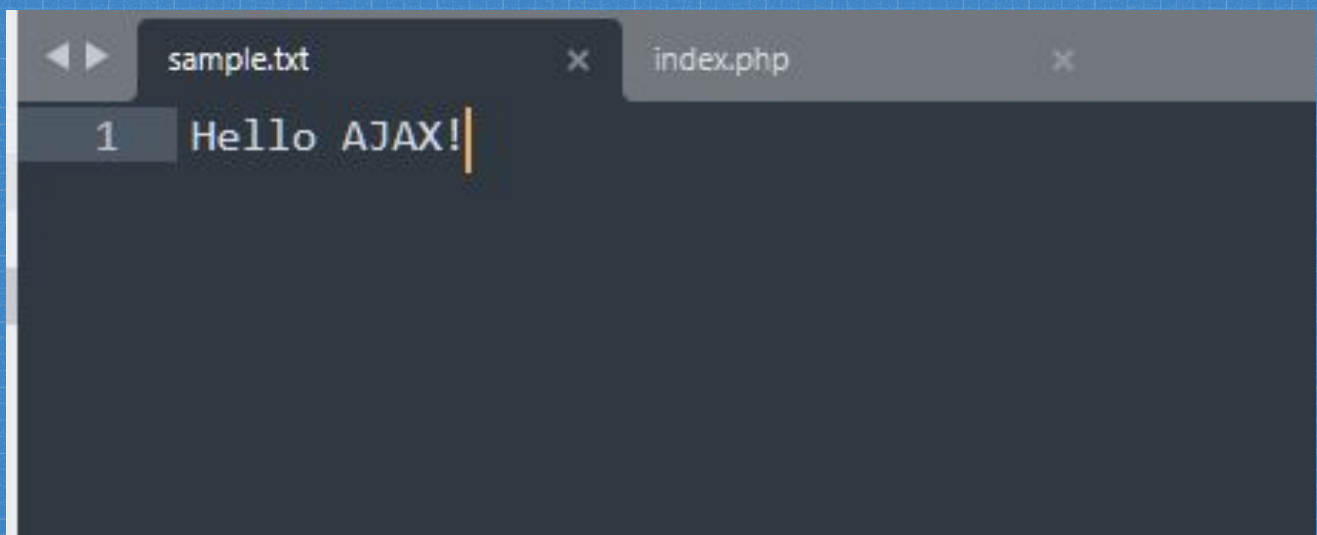
- Create new project inside htdocs



5

AJAX using Vanilla JavaScript (exercise)

- sample.txt



6

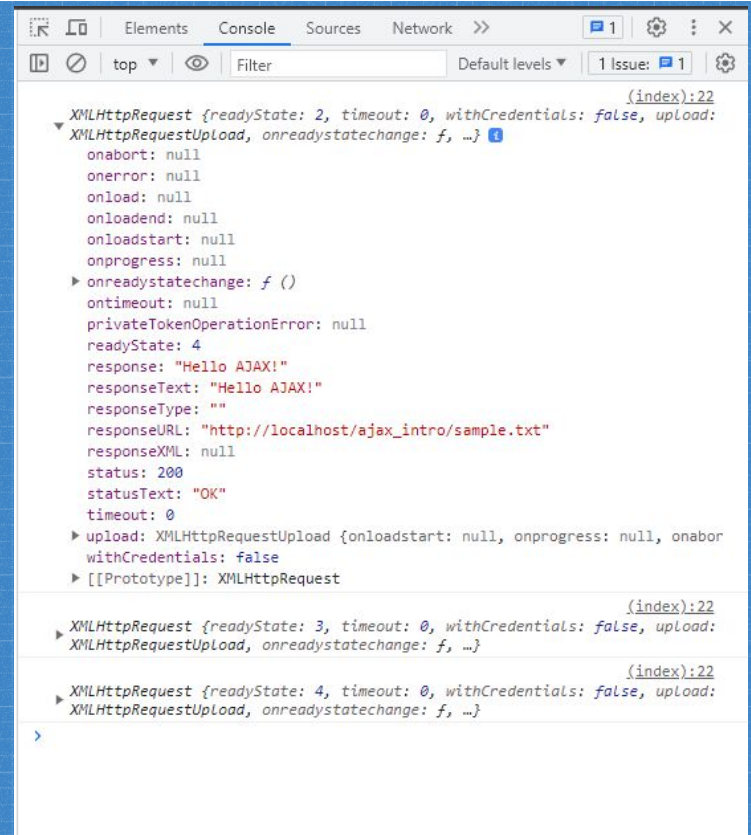

```
sample.txt x index.php x
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <meta charset="utf-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1">
6   <title></title>
7 </head>
8 <body>
9   <h1 id="demo">Change me!</h1>
10  <input type="text">
11  <button onclick="changeH1()">Change H1</button>
12 </body>
13 <script type="text/javascript">
14   function changeH1(){
15     var xhttp = new XMLHttpRequest();
16     xhttp.open("GET","sample.txt",true);
17     xhttp.send();
18     xhttp.onreadystatechange = function(){
19       if(this.readyState == 4 && this.status == 200){
20         document.getElementById("demo").innerHTML = this.responseText;
21       }
22       console.log(this);
23     };
24   }
25 </script>
26 </html>
27
```

AJAX using Vanilla JavaScript (exercise)



AJAX using Vanilla JavaScript (exercise)

cntrl+shift+i
//console.log()



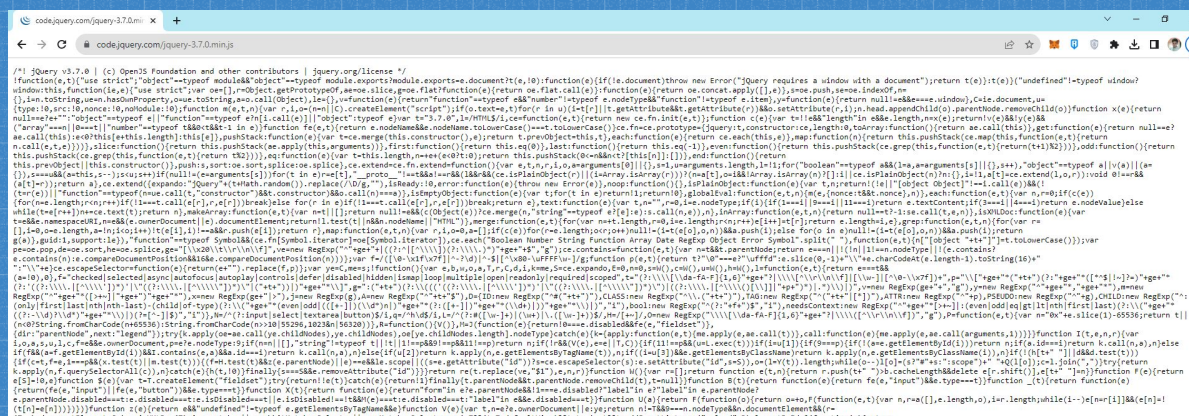
```
(index):22
XMLHttpRequest {readyState: 2, timeout: 0, withCredentials: false, upload:
XMLHttpRequestUpload, onreadystatechange: f, ...}
  abort: null
  onerror: null
  onload: null
  onloadend: null
  onloadstart: null
  onprogress: null
  ▶ onreadystatechange: f ()
  ontimeout: null
  privateTokenOperationError: null
  readyState: 4
  response: "Hello AJAX!"
  responseText: "Hello AJAX!"
  responseType: ""
  responseURL: "http://localhost/ajax_intro/sample.txt"
  responseXML: null
  status: 200
  statusText: "OK"
  timeout: 0
  ▶ upload: XMLHttpRequestUpload {onloadstart: null, onprogress: null, onabon
withCredentials: false
  ▶ [[Prototype]]: XMLHttpRequest

(index):22
XMLHttpRequest {readyState: 3, timeout: 0, withCredentials: false, upload:
XMLHttpRequestUpload, onreadystatechange: f, ...}

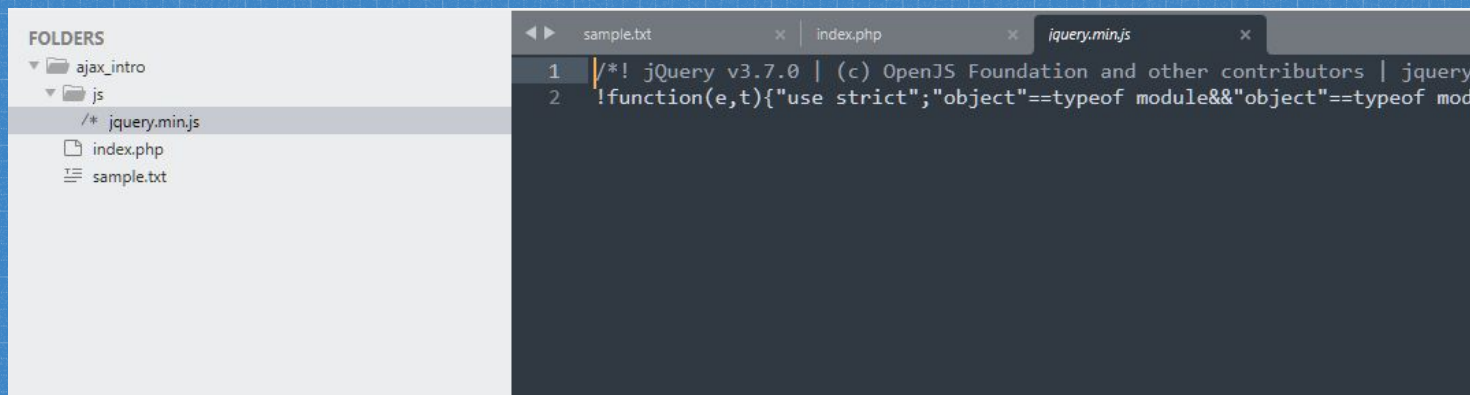
(index):22
XMLHttpRequest {readyState: 4, timeout: 0, withCredentials: false, upload:
XMLHttpRequestUpload, onreadystatechange: f, ...}
>
```

2 jQuery

- LINK:
 - <https://code.jquery.com/jquery-3.7.0.min.js>



- Copy and paste code in js/jquery.min.js



index.php

```
index.php x sample.txt x jquery.min.js x
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <meta charset="utf-8">
5     <meta name="viewport" content="width=device-width, initial-scale=1">
6     <title></title>
7     <script type="text/javascript" src="js/jquery.min.js"></script>
8 </head>
```

13

index.php

```
<script type="text/javascript">
    function changeH1(){
/*      var xhttp = new XMLHttpRequest();
        xhttp.open("GET","sample.txt",true);
        xhttp.send();
        xhttp.onreadystatechange = function(){
            if(this.readyState == 4 && this.status == 200){
                document.getElementById("demo").innerHTML = this.responseText;
            }
            console.log(this);
        };
*/
        $.get('sample.txt', function(data) {
            $('#demo').html(data);
        }, 'text');
    }

</script>
```

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AJAX with JQuery

- Use the activity “oop”

```
▼ OOP
  ▼ db
    🐘 db.php
    🐘 request.php
    🐘 index.php
    JS jquery.min.js
```

index.php

```
1  <?php
2      require "db/db.php";
3      $mydb = new myDB();
4      $mydb->select('tbl_students');
5      $student_data = $mydb->res;
6  ?>
7  <!DOCTYPE html>
8  <html lang="en">
9  <head>
10     <meta charset="UTF-8">
11     <meta name="viewport" content="initial-scale=1.0">
12     <title>Document</title>
13     <script type="text/javascript" src="jquery.min.js"></script>
14 </head>
15 <body>
16 >   <form id="addStudentForm" method="post" action="db/request.php">...
21   </form>
22   <table>
23 >     <thead>...
29     </thead>
30     <tbody id="tbodyStudent">
31
32     </tbody>
33   </table>
34 </body>
35 <script type="text/javascript">
36     $(document).ready(function(){
37         loadStudents();
38     })
39 >   function loadStudents(){ ...
66
67 >   $("#addStudentForm").on("submit",function(e){ ...
93 </script>
94 </html>
```


index.php // form

```
<form id="addStudentFomr" method="post" action="db/request.php">
  <input type="text" name="full_name">
  <input type="text" name="email">
  <input type="text" name="course_year_section">
  <input type="submit" name="add_student" value="ADD">
</form>
<table>
```

index.php // table body

```
<tbody id="tBodyStudent">
  |
</tbody>
table>
```


index.php

// script

```
function loadStudents(){
    $.ajax({
        url:"db/request.php",
        method:"POST",
        data:{
            "get_student":true,
        },
        success:function(result){
            //console.log(result)
            var tBody = ``;
            var cnt = 1;
            var datas = JSON.parse(result);
            datas.forEach(function(data){
                tBody +=`<tr>`;
                tBody +=`<td>${data['full_name']}
```

request.php

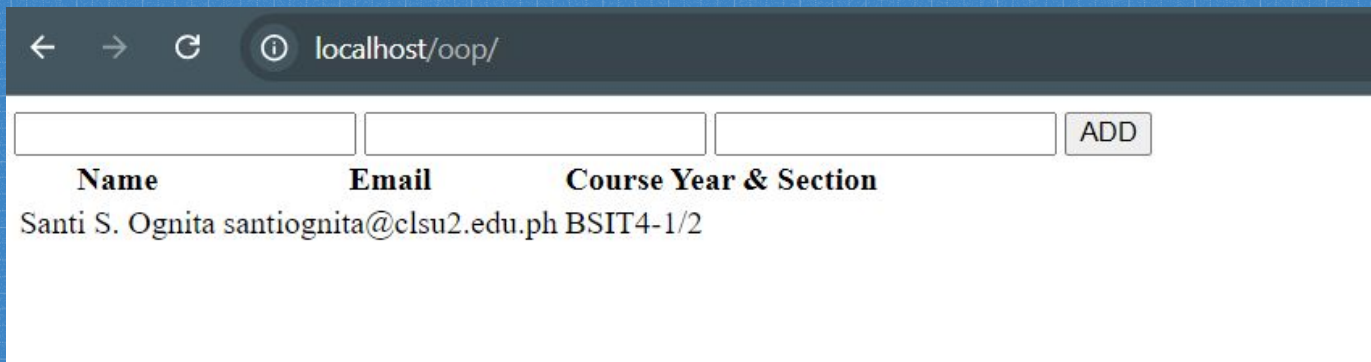
```
<?php

require "db.php";
$mydb = new myDB();

> if(isset($_POST['add_student'])){ ...
}

✓ if(isset($_POST['get_student'])){
    $mydb->select('tbl_students');
    $datas = [];
    ✓ while($row = $mydb->res->fetch_assoc()){
        array_push($datas,$row);
    }
    echo json_encode($datas);
} ?>
```


OUTPUT



localhost/oop/

Name	Email	Course Year & Section
Santi S. Ognita santiognita@clsu2.edu.ph BSIT4-1/2		

index.php

// script

```
$("#addStudentForm").on("submit",function(e){  
    e.preventDefault();  
    var datas = $(this).serializeArray();  
    var data_array = {};  
    $.map(datas,function(data){  
        data_array[data['name']] = data['value'];  
    });  
    console.log(datas);  
    console.log(data_array);  
})
```


OUTPUT

//console

```
▼ (3) [{...}, {...}, {...}] ⓘ oop/:68  
  ▶ 0: {name: 'full_name', value: 'a'}  
  ▶ 1: {name: 'email', value: 'a'}  
  ▶ 2: {name: 'course_year_section', value: 'a'}  
    length: 3  
  ▶ [[Prototype]]: Array(0)  
  
▼ {full_name: 'a', email: 'a', course_year_section: 'a'} ⓘ oop/:69  
  course_year_section: "a"  
  email: "a"  
  full_name: "a"  
  ▶ [[Prototype]]: Object
```

index.php

// script

```
$("#addStudentForm").on("submit",function(e){  
    e.preventDefault();  
    var datas = $(this).serializeArray();  
    var data_array = {};  
    $.map(datas,function(data){  
        data_array[data['name']] = data['value'];  
    });  
    //console.log(datas);  
    //console.log(data_array);  
    $.ajax({  
        url:"db/request.php",  
        method:"POST",  
        data:{  
            'add_student':true,  
            ...data_array,  
        },  
        success:function(result){  
            //console.log(result);  
            loadStudents();  
        },  
        error:function(error){  
            //console.log(error);  
            alert("Something went wrong!");  
        }  
    })  
})  
})
```


request.php

```
if(isset($_POST['add_student'])){  
    unset($_POST['add_student']);  
    $mydb->insert('tbl_students',$_POST);  
}
```

db.php

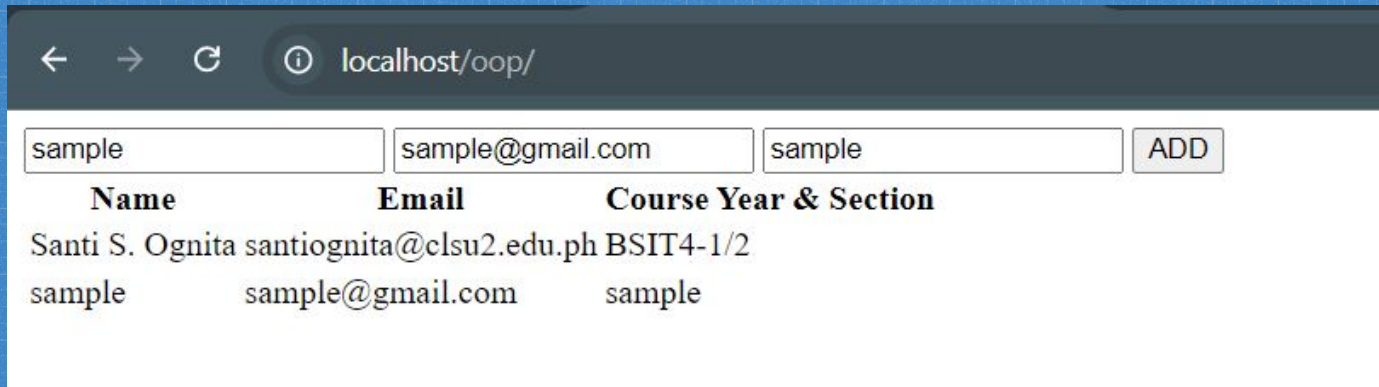
1 reference | 0 overrides

```
public function insert($table,$data){  
    try{ ...  
    }catch(Exception $e){  
        echo („Error while inserting data! <br>".$e);  
    }  
}
```

2 references | 0 overrides

```
public function select($table,$row="*", $where=NULL){  
    try{ ...  
    }catch(Exception $e){  
        echo („Error requesting data! <br>".$e);  
    }  
}
```


OUTPUT



The screenshot shows a web browser window with the address bar displaying 'localhost/oop/'. Below the address bar is a form with three input fields and an 'ADD' button. The first input field contains 'sample', the second contains 'sample@gmail.com', and the third contains 'sample'. Below the form is a table with three columns: 'Name', 'Email', and 'Course Year & Section'. The table contains two rows of data. The first row shows 'Santi S. Ognita', 'santiognita@clsu2.edu.ph', and 'BSIT4-1/2'. The second row shows 'sample', 'sample@gmail.com', and 'sample'.

Name	Email	Course Year & Section
Santi S. Ognita	santiognita@clsu2.edu.ph	BSIT4-1/2
sample	sample@gmail.com	sample

Activity 2

1. ADD UPDATE FUNCTION
2. ADD DELETE FUNCTION
3. ADD SEARCH FUNCTION ON TABLE DATA
(DON'T USE DATATABLES PLUG-IN or any PLUGINS)
4. ADD DESIGN (BOOTSTRAP OR CUSTOM CSS)